Final Validation

Finding	Validation Method	Result	Conclusion
1. Everyday is the largest customer segment.	Recount customers by segment and verify the distinct customer total.	Everyday: 711; Premium: 295; Student: 278; SME: 216. Total: 1,500.	Supported. The segment counts sum to the full customer dataset.
2. Transfers generate the highest total transaction value.	Recalculate transaction counts and total amounts by transaction type.	Transfers: 3,549 transactions and approximately NGN 237.92 million.	Supported by the reported results.
3. The Mobile App has the lowest success rate.	Recalculate successful transactions divided by total transactions for each channel.	Mobile App: 89.75%; ATM: 91.70%.	Supported by the reported channel results.
4. International transactions have a higher risk review rate.	Recalculate risk reviews divided by total transactions for each international status.	International: 36.88%; domestic: 18.88%.	Supported. The international rate is substantially higher, but this does not prove the cause of the difference.
5. Transfers have the highest risk review rate by transaction type.	Recalculate risk review rates for each transaction type and compare the results.	Transfers: 28.49%; cash withdrawals: 25.31%.	Supported by the reported results. Further investigation is needed to explain the difference.
6. SMEs have the highest value per active customer.	Recalculate total transaction value divided by active customers within each segment.	SME: approximately NGN 387,923; Student: approximately NGN 386,604.	Supported by the reported results. SME ranks highest on this measure.
7. Transaction activity recovered in March.	Recalculate monthly transaction counts and total values using the transaction date.	February: 3,734 transactions; March: 4,133. March total value was approximately NGN 196.20 million.	Supported by the reported results. The reason for the February decline remains unknown.

Additional validation checks

1. Transaction-to-customer relationship

- **Finding:** Transactions can be linked to the customer dataset.
- **Validation method:** Check whether every transaction Customer_ID exists in the customer table.
- **Result:** Week 2 assessment found zero unmatched customer IDs.

- **Conclusion:** The customer and transaction datasets have a valid relationship for the analyses performed, subject to confirming the same result in the final files.

2. Missing values and outliers

- **Finding:** The transaction dataset contains missing values and potential high-value outliers.
- **Validation method:** Recheck missing-value counts and recalculate the IQR threshold.
- **Result:** You identified 96 missing Device_Type values, 96 missing Location values, and 1,474 potential transaction amount outliers.
- **Conclusion:** These issues must remain documented. The outliers should not automatically be removed, and the missing values should be handled according to a documented decision.

Findings that changed after validation

- Customer segment performance: Everyday customers generated the highest total transaction value, but SMEs had the highest value per active customer. These findings measure different things and are not contradictory.
- International transaction risk: International transactions had a higher risk review rate, but domestic transactions accounted for more total risk reviews. Rate and count should not be confused.
- High-value transactions: The IQR threshold identifies potential outliers, not confirmed fraudulent transactions.
- Data quality: Missing device and location values were identified, but identification alone does not mean they have been resolved.